Answer Key

Physics			
Q. No.	**Answer**	**Topic Name**	**Chapter Name**
1	(2)	Electrostatic Potential	Electrostatics
2	(4)	Moment of Inertia	System of Particles and Rotational Motion
3	(1)	Young's Modulus	Mechanical Properties of Solids
4	(1)	Distribution of Charge on Conductor	Electrostatics
5	(4)	Latent Heat	Thermal Properties of Matter
6	(4)	Lens Maker's Formula	Ray Optics
7	(4)	Gauss's Law	Electrostatics
8	(2)	pn Diode	Semiconductor Electronics
9	(4)	Surface Charge Density	Electrostatics
10	(3)	Errors in Measurements	Units and Measurements
11	(3)	Specific Heat	Kinetic Theory of Gases
12	(4)	Combination of Lenses	Ray Optics
13	(2)	Stationary Waves in a String	Waves
14	(3)	Magnetic Field at the Centre of a Magnetic Loop	Moving Charges and Magnetism
15	(2)	2nd Law of Motion	Laws of Motion
16	(1)	Projectile Motion	Motion in a Plane
17	(4)	Superposition of Waves	Waves
18	(2)	YDSE	Wave Optics
19	(4)	Nuclei Density	Nuclei
20	(2)	Spring-Mass System	Laws of Motion
21	[2]	Surface Charge Density, De Broglie Wavelength	Electrostatics, Dual Nature of Radiation and Matter
22	[4]	Bulk Modulus	Mechanical Properties of Fluids
23	[15]	Capacitor with Dielectric	Electrostatics
24	[40]	Torque, Rotational Motion	System of Particles and Rotational Motion
25	[3]	Rotational Equilibrium	System of Particles and Rotational Motion

Chemistry			
Q. No.	**Answer**	**Topic Name**	**Chapter Name**
26	(4)	Chemical kinetics	Order of reaction
27	(4)	Some basic concepts of chemistry	Hybridisation
28	(1)	Nitrogen containing compound	Amides
29	(1)	Hydrocarbons and halogen derivatives	Chemical properties
30	(2)	Hydrocarbons and halogen derivatives	Chemical properties
31	(4)	Coordination compounds	Stereochemistry
32	(3)	Some basic concepts of chemistry	Physical properties
33	(2)	Solution	Roult's law
34	(1)	Solution	Depression in freezing point
35	(4)	Hydrocarbons and halogen derivatives	IUPAC nomenclature
36	(3)	Coordination compounds	Magnetic moment

37	(4)	Halogen derivatives and hydrocarbons	Chemical properties
38	(2)	Hydrocarbons	% of atoms
39	(1)	Ionic equilibria	pH
40	(2)	Some basic concepts of chemistry	Electronic configuration
41	(4)	Coordination compounds	Hybridisation
42	(1)	Oxygen containing compounds	Chemical properties
43	(3)	Oxygen containing compounds	Aldol condensation
44	(3)	Coordination compounds	Magnetic moment
45	(4)	Ionic equilibria	Nature of compounds
46	[98]	Thermochemistry	Enthalpy
47	[54]	Electrochemistry	Bohr's atomic model
48	[1]	Some basic concepts of chemistry	Molarity
49	[5]	Ionic equilibria	Equilibrium constant
50	[10]	Cell reaction	Electrochemistry

Mathematics			
Q. No.	**Answer**	**Topic Name**	**Chapter Name**
51	(1)	Shortest distance	3D
52	(2)	Properties of matrix	Matrix
53	(4)	Monotonic function	Application of derivative
54	(1)	Conditional probability	Probability
55	(3)	Special series	Sequence and series
56	(2)	Roots	Quadratic equation
57	(1)	Slope of a line	Straight line
58	(4)	Properties of definite integration	Definite integration
59	(4)	Normal vector	Vector
60	(3)	Focus	Ellipse
61	(3)	By parts	Definite integration
62	(1)	Rotation of line	Straight line
63	(4)	No. of triangles	Permutation and combination
64	(1)	Real part of complex number	Complex number
65	(3)	Types of relation	Set and relation
66	(4)	Rational terms	Binomial thorem
67	(3)	First order differential equation	Differential equation
68	(1)	Basic identities	Inverse trigonometry
69	(2)	Properties of adjoint	Matrix
70	(4)	Limit with series expansion	Limit
71	[15]	Area between curves	Area
72	[96]	Domain	Function
73	[56]	Area of triangle	3D
74	[63]	Expansion	Binomial theorem
75	[30]	Contact between two curve	Parabola

Answers with Explanation

Physics

1. Option (2) is correct.

In a conducting spherical shell of radius R, for all the points inside it, $V = \frac{kq}{R}$ =constant.

Work done in moving any charge from a point A to another point B, inside the shell is given by $W_{AB} = q(V_B - V_A)$

Since, $V_A = V_B$, $W_{AB} = 0$

2. Option (4) is correct.

Moment of inertia of a ring about a diameter is given by $I = \frac{MR^2}{2}$

Here, $M = \lambda L$

and $L = 2\pi R$

$\Rightarrow \quad R = \frac{L}{2\pi}$

$\therefore \quad I = \frac{(\lambda L)}{2} \times \left(\frac{L}{2\pi}\right)^2$

$= \frac{\lambda L^3}{8\pi^2}$

3. Option (1) is correct.

Young's Modulus $= \frac{F}{A} \times \frac{l}{\Delta l}$

$F = Mg$

$= 50 \times 3\pi$

$= 150\pi$ (N)

$A = \pi r^2$

$= \pi(3 \times 10^{-3})^2$

$= 9\pi \times 10^{-6}\ m^2$

$l = 3$ m

$\Delta l = 0.1 \times 10^{-3}$m

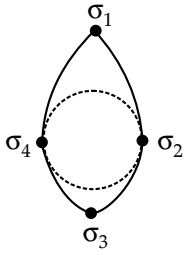
$\therefore \quad Y = \frac{150\pi \times 3}{9\pi \times 10^{-6} \times 0.1 \times 10^{-3}}$

$= 5 \times 10^{11}\ N/m^2$

$= P \times 10^{11}\ N/m^2$

On comparing, $P = 5$.

4. Option (1) is correct.



Every point on a metallic surface has the same potential $V = \frac{Kq}{r}$ where r is the radius of curvature at that point. Also, $\sigma \propto \frac{q}{r^2} \Rightarrow \sigma \propto \frac{q}{r}$ as $q \propto r$. In the given figure, $r_1 < r_3 < (r_2 = r_4)$

Hence, $\sigma_1 > \sigma_3 > (\sigma_2 = \sigma_4)$

5. Option (4) is correct.

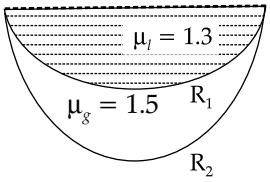
Using conservation of energy

$Mgh = Ms\Delta T$

$\Delta T = \frac{gh}{s} = \frac{10 \times 200}{4200}$

$\Delta T = 0.48$ K.

6. Option (4) is correct.



The equivalent focal length is given by

$\frac{1}{f} = \frac{1}{f_{lg}} + \frac{1}{f_{gl}}$

$\frac{1}{f_{lg}} = (\mu_l - 1)\left(\frac{1}{\infty} - \frac{1}{-R_1}\right)$

$= 0.3 \times \frac{1}{30} = \frac{1}{100}$

$\frac{1}{f_g l} = (\mu_g - 1)\left(\frac{1}{-R_1} - \frac{1}{-R_2}\right)$

$= 0.5\left(\frac{1}{-30} - \frac{1}{-20}\right)$

$= 0.5 \times \frac{1}{60}$

$= \frac{1}{120}$

$$\frac{1}{f} = \frac{1}{100}+\frac{1}{120} = \frac{11}{600}$$

$$f = \frac{600}{11}\text{cm}$$

7. Option (4) is correct.

In a cube the maximum distance between two points is $\sqrt{3}l$ (body diagonal).

From Gauss's theorem,

$$\phi_{MX} = \frac{q_{MX}}{\varepsilon_0} = \frac{\lambda(\sqrt{3}l)}{\varepsilon_0}$$

As,

$$\frac{1}{\varepsilon_0} = 4\pi \times 9 \times 10^9$$

$$\therefore \quad \phi_{MX} = \lambda(\sqrt{3}l)\times 4\pi\times 9\times 10^9$$

$$= 2\times 10^{-9}\times\sqrt{3}\times(\sqrt{3}\times 10^{-2})\times 36\pi\times 10^9$$

$$= 2.16\pi \text{ Nm}^2/\text{C}$$

$$= x \text{ Nm}^2/\text{C}$$

On comparing, we get $x = 2.16\pi$

8. Option (2) is correct.

The given circuit is the implementation of AND gate. If A = 1 and B = 0, output C = 0. Here, 5V is represented by symbol 1 and 0 V or ground by the symbol 0. Hence, the output is $V_0 = 0$

9. Option (4) is correct.

Let q_1 and q_2 be the initial charges on spheres with radii R and 3R respectively.

$q_{sys} = q_1 + q_2 = 4\pi\sigma(R^2 + 9R^2) = 40\pi\sigma R^2$

When they are brought in contact, both acquire a common potential V. Let the final charges on both be q'_1 and q'_2. respectively.

Since, charge is conserved for the system

$$q'_1 + q'_2 = q_1 + q_2 \quad \text{...(i)}$$

Common potential $V^1 = \frac{Kq'_1}{R} = \frac{Kq'_2}{3R}$

$$q'_2 = 3q'_1 \quad \text{...(ii)}$$

From (i) and (ii), we get

$$4q'_1 = 40\pi\sigma R^2$$

$$q'_1 = 10\pi\sigma R^2$$

and $q'_2 = 30\pi\sigma R^2$

Now, $\frac{\sigma_1}{\sigma_2} = \frac{q'_1}{R^2}\times\frac{9R^2}{q'_2}$

$$= \frac{9q'_1}{q'_2} = 9\times\frac{1}{3} = 3$$

10. Option (3) is correct.

$$Q = X^2Y^{3/2}Z^{-2/5} = \frac{Y^{3/2}}{X^2Z^{2/5}}$$

$$\frac{\Delta Q}{Q} = \frac{3}{2}\frac{\Delta Y}{Y}+2\frac{\Delta X}{X}+\frac{2}{5}\frac{\Delta Z}{Z}$$

(For maximum error)

$$= \left(\frac{3}{2}\times 0.2\right)+2(0.1)+\left(\frac{2}{5}\times 0.5\right)$$

$$= 0.3 + 0.2 + 0.2$$

$$= 0.7$$

11. Option (3) is correct.

Sudden compression implies adiabatic process.

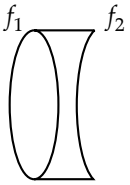
Here, $P_1V_1^{\gamma} = P_1V_1^{\gamma}$

Given, $V_2 = \frac{V_1}{8}$

$$\therefore \quad \frac{p_2}{p_1} = \left(\frac{V_1}{V_2}\right)^{\gamma}$$

$$= (8)^{5/3} = 2^5 = 32$$

12. Option (4) is correct.



The equivalent focal length of the combination is given by

$$\frac{1}{f} = \frac{1}{f_1}+\frac{1}{f_2} = \frac{1}{30}+\frac{1}{-20}$$

$$= \frac{2-3}{60}$$

$$= -\frac{1}{60}$$

–ve sign implies that the system behaves as diverging lens.

From lens formula

$$\frac{1}{v}-\frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v}-\frac{1}{-20} = \frac{1}{-60}$$

$$\frac{1}{v} = -\frac{1}{20}-\frac{1}{60}$$

$$= \frac{-4}{60} = -\frac{1}{15}$$

v = –15 cm (On the left of the lens combination given)

$|v|$ = 15 cm

13. Option (2) is correct.

Velocity of a transverse wave in a string is given by $v = \sqrt{\frac{T}{\pi r^2 \rho}}$

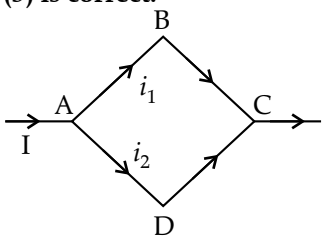
Since, both strings are made of the same material ρ is same for both.

Hence, $v \propto \frac{1}{r}$

$$\frac{v_2}{v_1} = \frac{r_1}{r_2}$$

$$= \frac{R}{\frac{R}{2}} = 2$$

14. Option (3) is correct.



Given $R_{ABC} = r$

$R_{ADC} = 2r$

Since, R_{ABC} is parallel with R_{ADC}

$$\frac{2r}{r} = \frac{i_1}{i_2} = 2$$

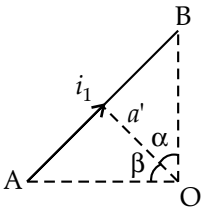
$$i_1 = 2i_2$$

and $i_1 + i_2 = I$

$\Rightarrow$ $i_2 = \frac{I}{3}$ and $i_1 = \frac{2I}{3}$

$$B_0 = \frac{\mu_0 i_1}{4\pi a'}(\sin\alpha + \sin\beta)$$

Here, $\alpha = \beta = \frac{\pi}{4}$ and $a' = \frac{a}{2}$



$\therefore$ $B_{0(AB)} = \frac{\mu_0\left(\frac{2I}{3}\right)}{4\pi\left(\frac{a}{2}\right)} \times \left(2 \times \frac{1}{\sqrt{2}}\right)$

$$= \frac{\mu_0 I\sqrt{2}}{3\pi a}$$

Also, $B_{0(BC)} = B_{0(AB)}$

$\therefore$ $B_{0(BC)} + B_{0(AB)} = 2 \times \frac{\mu_0 I\sqrt{2}}{3\pi a}$

$$= \frac{2}{3}\frac{\mu_0 I\sqrt{2}}{\pi a}$$

Likewise $B_{0(AD)} + B_{0(DC)} = 2 \times \frac{u_0 I\sqrt{2}}{6\pi a}$

$$= \frac{u_0 I\sqrt{2}}{3\pi a}$$

$\therefore$ $B_0 = \frac{\sqrt{2}\mu_0 I}{3\pi a}$

15. Option (2) is correct.

Since, $\vec{F} = 6\hat{k}$, being along Z-axis, cannot affect v_x and v_y components, which hence, forth remains same. But body acquires another component.

$$v_z = a_z t$$

Here $a_z = \frac{F_z}{m} = \frac{6}{2} = 3\ \text{m/s}^2$

and $t = \frac{5}{3}s$

$$v_z = 3 \times \frac{5}{3} = 5\ \text{m/s}$$

$$\vec{v}\left(t = \frac{5}{3}\text{s}\right) = 3\hat{i} + 4\hat{j} + 5\hat{k}$$

16. Option (1) is correct.

Given , $H_1 = 8H_2$

$$\frac{u^2 \sin^2\theta_1}{2g} = \frac{u^2 \sin^2\theta_2}{2g} \times 8$$

$$\sin\theta_1 = 2\sqrt{2}\sin\theta_2$$

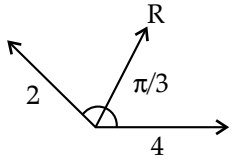
$$\frac{T_1}{T_2} = \frac{\sin\theta_1}{\sin\theta_2} = 2\sqrt{2} : 1$$

17. Option (4) is correct.

Given, $y_1 = 4\sin(kz - \omega t)$

$$y_2 = 2\sin(kx - \omega t + \frac{2\pi}{3})$$

Using phasor diagram,



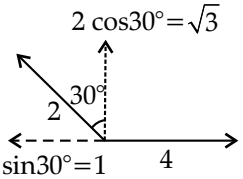
Using phasor diagram

$$R = \sqrt{4 + 16 + 16\cos\frac{2\pi}{3}}$$

$$I = \sqrt{20 - 16 \times \frac{1}{2}}$$

$$= \sqrt{12} = 2\sqrt{3}$$

$$R_x = 4 - 1 = 3$$



$$R_Y = \sqrt{3}$$

$$\tan\theta = \frac{R_Y}{R_X}$$

$$= \frac{1}{\sqrt{3}}$$

$$\theta = \frac{\pi}{6}$$

18. Option (2) is correct.

The primary requirement for the two waves to interfere is that both should have same frequency or wavelength.

Since, $\lambda_R > \lambda_G$, there shall be no interference fringes.

19. Option (4) is correct.

Mass density of nucleus $\rho = \frac{M}{V} = \frac{Am_p}{\frac{4}{3}\pi[R_0 A^{\frac{1}{3}}]^3}$

$$= \frac{m_p}{\frac{4}{3}\pi R_0^3}$$

$$= \text{constant}$$

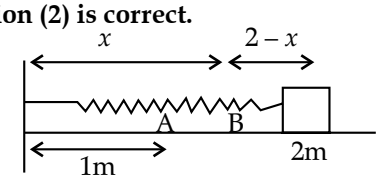
Here, A = mass number

m_p = mass of proton

R_0 = constant

Hence, mass density of nucleus is independent of its mass number A.

20. Option (2) is correct.



Block is released from position A. where spring is compressed through 1 m. COE at positions A and B gives

$$\frac{1}{2}K(1)^2 = \frac{1}{2}K(2-x)^2 + \frac{1}{2}mv^2$$

$$v^2 = \frac{K}{m}[1^2 - (2-x)^2]$$

$$= 100[1^2 - (2-x)^2]$$

$$\therefore \quad v = 10[1-(2-x)^2]^{\frac{1}{2}}$$

21. Correct answer is [–2].

De Broglie wavelength $\lambda = \frac{h}{mv}$

$$v = u + at$$

Since, electron is released from rest, $u = 0$

$$v = at = \frac{eEt}{m}$$

$$= \frac{e}{m}\frac{\sigma}{2\varepsilon_0}t$$

$$\therefore \quad \lambda = \frac{h}{\frac{m.e\sigma t}{2m\varepsilon_0}}$$

$$= \frac{2\varepsilon_0 h}{e\sigma t}$$

$$\frac{d\lambda}{dt} = \frac{-2\varepsilon_0 h}{e\sigma t^2}$$

$$\frac{d\lambda}{dt} \propto t^{-2}$$

So, $n = -2$

22. Correct answer is [4].

Bulk Modulus $= B = \frac{\Delta P}{\frac{\Delta V}{V}}$

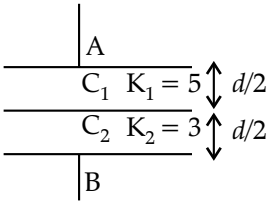
$$\frac{\Delta V}{V} = \frac{\Delta P}{B}$$

$$V = \frac{B\Delta V}{\Delta p}$$

$$= \frac{2\times10^9 \times 0.8\times10^{-6}}{(5-1)\times10^5}$$

$$= 4 \times 10^{-3}\ m^3 = 4 \text{ litres}$$

23. Correct answer is [15].



$$C_1 = K_1\frac{\varepsilon_0 A}{\frac{d}{2}} = 10\frac{\varepsilon_0 A}{d}$$

$$C_2 = K_2\frac{\varepsilon_0 A}{\frac{d}{2}} = 6\frac{\varepsilon_0 A}{d}$$

$$C_{AB} = \frac{C_1 C_2}{C_1 + C_2}$$

$$= \frac{\varepsilon_0 A}{d}\left(\frac{10\times6}{16}\right)$$

$$= \frac{15}{4}\frac{\varepsilon_0 A}{d}$$

$$C_{AB} = \frac{15}{4}\times\frac{8.85\times10^{-12}\times4\times10^{-4}}{1.77\times10^{-3}}$$

$$= \frac{15}{4}\times20\times10^{-13}\,F$$

$$= 7.5\ pF$$

When another capacitor of C = 7.5 pF is connected in parallel to C_{AB}, effective capacitance becomes

$$C' = C_{AB} + 7.5$$
$$= 7.5 + 7.5$$
$$= 15\ pF$$

24. Correct answer is [40].

Torque $\tau = I\alpha$

$$\alpha = \frac{25\pi}{I}$$

Where I = moment of inertia of the disc about the diameter

$$\omega_0 = 1800\times\frac{2\pi}{60} = 60\pi\,(rad/s)$$

$$\omega = 2100\times\frac{2\pi}{60} = 70\pi\ (rad/s)$$

$$t = 40\ s$$

Applying $\omega = \omega_0 + \alpha t$

$$70\pi = 60\pi + \frac{25\pi}{I}\times40$$

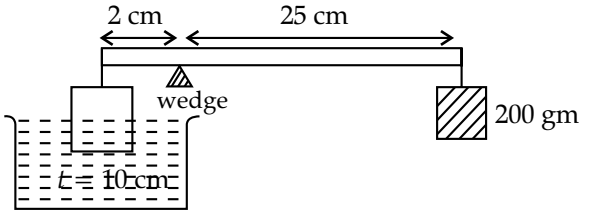
$$10 = \frac{1000}{I}$$

$$I = 100 = \frac{MR^2}{4}$$

$$R = 20\ m$$

$$Diameter = 2R = 40\ m$$

25. Correct answer is [3].



Since, half of the volume of the cube is submerged effective weight of the cube

$$= Mg - \text{buoyant force}$$

$$= Mg - \frac{V}{2}\rho wg$$

Taking moments about the wedge, at equilibrium,

$$200\ g \times 25 = (Mg - 500g)\times 2$$

(In CGS system)

$$\frac{5000}{2} = M - 500$$

$$M = 3000\ g$$
$$= 3\ kg$$

Chemistry

26. Option (4) is correct.

Let the initial concentration be $[A]_0$

When the concentration is ¼ of the initial,

$$[A]_t = \frac{1}{4[A]_0}$$

Using the integrated rate law:

$$\ln\left(\frac{1}{4[A]_0}\right) - \ln[A]_0 = -kt_1$$

$$\ln\left(\frac{1}{4}\right) = -kt_1$$

$$t_1 = \left[\frac{In\left(\frac{1}{4}\right)}{-k}\right] = \ln\frac{(4)}{k}$$

when the concentration is 1/8 of the initial,

$$[A]_t = \frac{1}{8[A]_0}$$

Using the integrated rate law:

$$\ln\left(\frac{1}{8[A]_0}\right) - \ln[A]_0 = -kt_2$$

$$\ln\left(\frac{1}{8}\right) = -kt_2$$

$$t_2 = \frac{In\left(\frac{1}{8}\right)}{-k} = \ln\frac{8}{k}$$

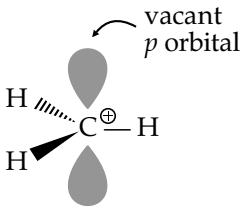
$$\frac{t_1}{t_2} = \frac{In\frac{(4)}{k}}{In\frac{(8)}{k}}$$

$$\frac{t_1}{t_2} = \frac{In4}{In8} = \frac{In2^2}{In2^3}$$

$$= \frac{2In2}{3In2}$$

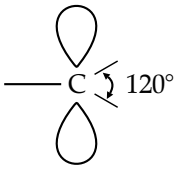
$$= \frac{2}{3}$$

27. Option (4) is correct.



Carbocation – sp^2 hybridised carbon with empty *p* orbital.

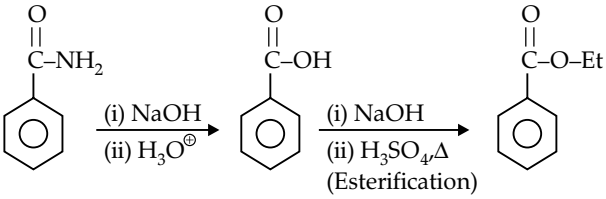
Carbon free radical– sp^2 or sp^3 hybridised a carbon with one unpaired electron.



Nucleophile– the species that can supply electron pair.

Electrophile – the species that can receive electron pair.

28. Option (1) is correct.

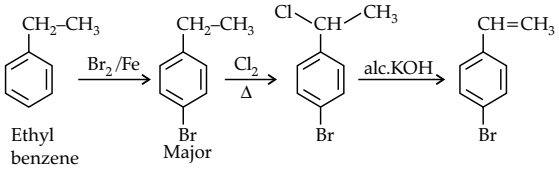


A - Benzamide shows positive Lassaigne's test.

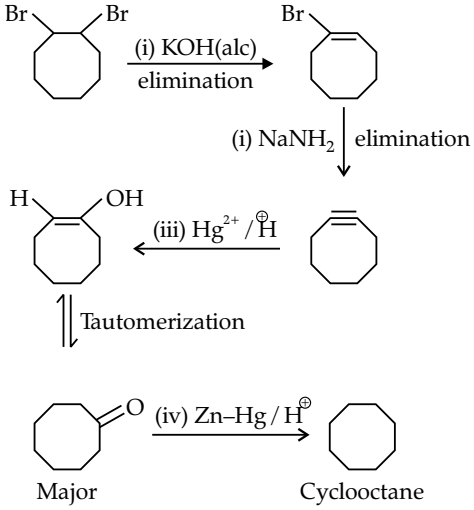
B - Benzoic acid gives effervescence with aq. $NaHCO_3$.

C - Ester gives fruity smell.

29. Option (1) is correct.



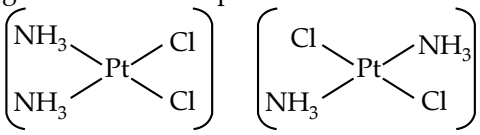
30. Option (2) is correct.



31. Option (4) is correct.

Statement I is correct statement. A homoleptic octahedral complex formed using monodentate ligands will not exhibit stereoisomerism. This is because the symmetry of the complex prevents the existence of different spatial arrangements for the identical ligands.

Statement II is incorrect. *Cis*- and *trans*-platin are NOT heteroleptic complexes of Pd. They are actually heteroleptic complexes of Pt. Specifically, they are square planar complexes where the central metal ion is platinum (Pt), and they contain two different types of ligands, ammonia and chloride. The prefix "cis" and "trans" indicates the spatial arrangement of the ligands around the platinum atom.



Cis-platin *trans*-platin

32. Option (3) is correct.

Atomic number 32 = Ge

Outermost E. C. = $4s^2,4p^2$

Atomic number 35 = Br

Outermost E. C. = $4s^2,4p^5$

Atomic number 87 = Fr

Outermost E. C. = $7s^1$

Atomic number 19 = K

Outermost E. C. = $4s^1$

Ionisation enthalpy is known as ionization energy, is the energy required to remove an electron from a gaseous atom in its ground state, transforming it into a positive ion.

Fr and K have same number of electrons in outermost shell which are less than others so their ionisation enthalpies are also low.

But out of Fr and K the atomic size of K is smaller as compared to the atomic size of Fr. So, ionization enthalpy of K is higher than Fr.

Hence, lowest first ionization enthalpy among the given element will be of Fr.

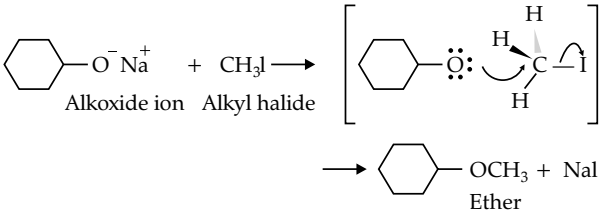
A binary mixture of C_6H_5OH and $C_6H_5NH_2$ exhibits negative deviation from Raoult's law. The vapour pressure of the solution is lower than the vapor pressures of the pure components. The boiling point of the solution is higher than the boiling points of the pure substances. Therefore, this mixture forms a maximum boiling azeotrope.

33. Option (2) is correct.

Binary mixture of C_6H_5OH and $C_6H_5NH_2$ shows negative deviation from Raoult's law

So vapour pressure of solution is less than vapour pressure of pure C_6H_5OH and $C_6H_5NH_2$

So boiling point of solution is greater than boiling point of pure C_6H_5OH and $C_6H_5NH_2$ so shows maximum boiling azeotrope.



34. Option (1) is correct.

$$\Delta T = i \times K_f$$
$$0.2 = i \times 1.8 \times 0.1$$
$$i = \frac{0.2}{0.18}$$
$$= \frac{20}{18}$$
$$= \frac{10}{9}$$

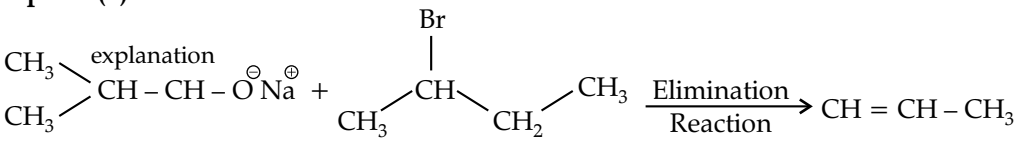
For $HA \leftrightarrow H^+ + A^-$

$t=0$ 1 0 0

$t=t$ $1-\alpha$ α α

$$i = 1 + \alpha$$
$$\frac{10}{9} = 1 + \alpha$$
$$\alpha = (10/9) - 1$$
$$= \frac{1}{9}$$
$$K_{eq} = [H^+][A^-]/[HA]$$
$$= Ca^2/(1-\alpha)$$
$$= 0.1\,(1/9)^2/1-(1/9)$$
$$= 1/720$$
$$K_{eq} = 1.38 \times 10^{-3}$$

35. Option (2) is correct.



36. Option (2) is correct.

In the combustion of organic compound, all "C" in CO_2 and all "H" in H_2O comes from organic compound.

$C_xH_yO_z + O_2 \rightarrow CO_2 + H_2O$

$$\text{Weight of carbon in } CO_2 = \left(\frac{12}{44}\right) \times 0.307$$
$$= 0.0838 \text{ gm}$$
$$\text{Weight of Hydrogen in } H_2O = \left(\frac{2}{18}\right) \times 0.127$$
$$= 0.0141 \text{ gm}$$

% of Hydrogen in compound = $(0.0141 \times 0.21) \times 100$

$$= 6.719\%$$
$$= 6.72\%$$

Weight of oxygen in compound = $0.210 - (0.0837 - 0.0141)$

$$= 0.1122$$

$$\%\text{ of oxygen in compound} = \left(\frac{0.1122}{0.21}\right) \times 100$$
$$= 53.41\%$$

37. Option (4) is correct.

4 – Ethylclopent –2 –en –1 –ol

38. Option (3) is correct.

Cu^+ :[Ar] $3d^{10}$ no unpaired electron Spin only magnetic moment = $\sqrt{n(n+2)}$ 0B.M. where n= no. of unpaired electrons

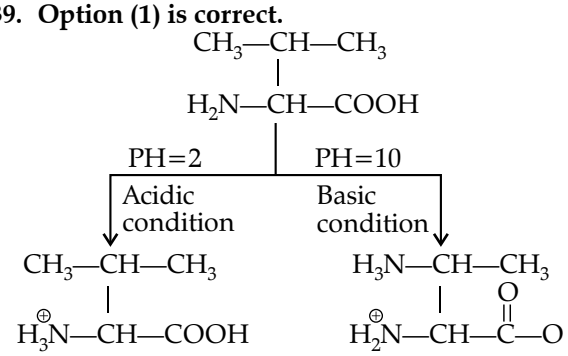
Cu^{+2} :[Ar] $3d^9$ one unpaired electron

Cr^{+2} : [Ar] $3d^4$ 4 unpaired electrons

Cr^{+3} : [Ar] $3d^3$ 3 unpaired electrons

Order of μ : $Cr^{+2} > Cr^{+3} > Cu^{+2} > Cu^+$

39. Option (1) is correct.



40. Option (2) is correct.

Element with atomic number 9 is fluorine.

The electronic configuration of a fluorine is $1s^2, 2s^2, 2p^5$.

A. 5 electrons can be up spin i.e. $m_s = +\frac{1}{2}$.

4 electrons can be down spin i.e. $m_s = -1/2$

Hence, it is correct statement.

B. There are 5 electrons in p orbit 2 in paired form in p_x, 2 in paired form in p_y and 1 in unpaired form in p_z. Hence, it is incorrect statement.

C. Last electron is in $2p_z$ hence, $n = 2$ and $l = 1$. Hence, it is correct statement.

D. Angular node for s orbital is 0 and for each p orbital is one so total angular nodes are 3. Hence, it is incorrect statement.

41. Option (3) is correct.

In $[Co(NH_3]_6]^{3+}$,

Electronic configuration of $Co^{+3}$is $[Ar]3d^6$

NH_3 is strong field ligand.

Hybridisation state of Co^{3+} is d^2sp^3

In SF_6 , Hybridisation state of sulphur is sp^3d^2

In $[CrF_6]^{3-}$

Electronic configuration of Cr^{+3} is $[Ar]3d^3$

Hybridisation state of Cr^{3+} is d^2sp^3

In $[CoF_6]^{3-}$

Electronic configuration of Co^{+3} is $[Ar]3d^6$

F^- is weak field ligand.

Hybridisation state of Co^{3+} is sp^3d^2

In $[Mn(CN)_6]^{3-}$

Electronic configuration of Mn^{+3} is $[Ar]3d^4$

CN^- is strong field ligand

Hybridisation state of Mn^{3+} is d^2sp^3

In $[MnCl_6]^{3-}$

Electronic configuration of Mn^{+3} is $[Ar]3d^4$

Cl^- is weak field ligand.

Hybridisation state of Cl^- is sp^3d^2

Total number of sp^3d^2 hybridised molecules are 3.

42. Option (1) is correct.

(1) Carboxylic acid gives effervescence with sodium bicarbonate solution.

(2) Phenolic-OH gives voilet coloured complex with neutral $FeCl_3$.

(3) Alcoholic-OH gives Red colour with ceric ammonium Nitrate.

(4) When alkaline $KMnO_4$ reacts with an unsaturated compound (Alkene or alkyne) the purple colour of $KMnO_4$ solution disappears, indicating positive test for unsaturation.

43. Option (3) is correct.

The given compound is a **1, 6-dicarbonyl compound** (specifically, a keto-aldehyde), and the question asks about **intramolecular aldol condensation**, which typically forms a **5- or 6-membered ring** due to favourable ring strain considerations.

Compound:

$O{=}CH{-}(CH_2)_4{-}CO{-}CH_3$

Numbering the carbon chain for clarity:

$OHC{-}CH_2{-}CH_2{-}CH_2{-}CH_2{-}CO{-}CH_2$

$C_1 \quad C_2 \quad C_3 \quad C_4 \quad C_5 \quad C_6 \quad C_7$

We have:

- An aldehyde group at one end (C_1)
- A ketone group at C_6

Intramolecular Aldol Condensation:

- Enolate formation happens at the α-carbon next to the ketone (i.e., C_5)
- The enolate attacks the **aldehyde carbonyl** at C_1

This forms a **new C–C bond between C_5 and C_1,** leading to a **5-membered ring.**

After Aldol Addition:

We get a **β-hydroxy ketone** (5-membered ring with an OH and a ketone)

Upon heating (condensation), this **dehydrates** (loss of H_2O) to form a **conjugated enone** system.

A **cyclopentene ring** with a ketone group and a double bond — this matches

44. Option (3) is correct.

In $[NiCO_4]$

Electronic configuration of Ni is $[Ar]3d^8,4s^2$

Shape – Tetrahedral

Number of unpaired electrons = 0

Magnetic Moment= 0

In $[Ni(CN)_4]^{-2}$

Electronic configuration of Ni^{+2} is $[Ar]3d^8,4s^0$

Shape – Square planar.

Number of unpaired electrons = 0

Magnetic Moment = 0

In $[NiCl_4]^{2-}$

Shape – Tetrahedral

Electronic configuration of Ni^{+2}[Ar] $3d^8,4s^2$.

Number of unpaired electrons = 2

Magnetic moment = 2.8 B. M

In $[MnBr_4]^{2-}$

Shape –Tetrahedral

Electronic configuration of Mn^{+2} is $[Ar]3d^5$

Number of unpaired electrons = 5

Magnetic moment = 5.9 B. M.

45. Option (4) is correct.

Statement - I The statement that H_2Se is more acidic than H_2Te is incorrect. In fact, H_2Te is the more acidic of the two. The acidity of hydrogen hydrides increases down the group due to decreasing bond dissociation enthalpy.

Statement – II Statement I is correct. H_2Se is less acidic than H_2Te as bond dissociation enthalpy of H_2Te is lower then H_2Se and hence, H^+ is dissociated with more ease.

46. Correct **answer is [98].**

$$\Delta H_{R.E} = \Delta H_{f(exp)} - \Delta H_{f(theo)}$$

$\Delta Hf_{(exp)}$ for $X_2\,Y_{(g)} = 80$ kJ/mole

for $\Delta H_{f(Theo)}$

$$X_{2(g)} + \frac{1}{2}Y_{2(g)} \rightarrow X_2Y_{(g)}\Delta H = ?$$

$$\Delta H_{f(Theo)}$$

$$= \left(BEx \equiv x + \frac{1}{2}BE_{y=y}\right) - (BE_{x=x} + BE_{x=y})$$

$$= \left(940 + \frac{1}{2}\times 500\right) - (410 + 602)$$

$$= 178 \text{ kJ/mole}$$

$$\Delta H_{R.E} = 80 - 178$$

$$= -98 \text{ kJ/mol}$$

$$|\Delta H_{R.E}| = 98$$

47. Correct answer is [54].

$$E_T = 13.6\frac{z^2}{n^2}ev$$

For energy of H-atom, energy of 1st

$E_1 = -13.6$ eV [$z = 1$, $n = 1$]

For Be^{+3} ion, energy of 1st E.S.

$$\frac{E_H}{E_{Be^{+3}}} = \frac{z_1^2}{n_1^2}\times\frac{n_2^2}{z_2^2}$$

$$\frac{E_H}{E_{Be^{+3}}} = \frac{1}{1}\times\frac{4}{16}$$

$$\Rightarrow \quad \frac{E_H}{E_{Be^{+3}}} = -13.6\times 4$$

$$|E_{Be^{+3}}| = 54.4\text{eV}$$

48. Correct answer is [1].

$NaI\,(aq) + AgNO_3\,(aq) \rightarrow AgI(s) + NaNO_3(aq)$

20 mL

$$\text{Moles of } I^- \text{ in NaI} = \text{Moles of } I^- \text{ in AgI} = \frac{4.74}{235}$$

$$\text{Moles of NaI} = \frac{4.74}{235}$$

$$\text{Molarity [NaI]} = \frac{4.74}{235} = 1.008$$

49. Correct answer is [5].

$H_2O(g) \leftrightarrow H_2(g) + \frac{1}{2}O_2(g)$

At $t = 0$ 1 mole

At $t = t$ eq $1-\alpha$ α $\alpha/2$

$$n_T = 1 + \frac{\alpha}{2} \simeq 1(\alpha << 1)$$

$$K_p = \frac{P_{H_2}.PO_2^{1/2}}{P_{H_2O}}$$

$$= \frac{(\alpha.p)\left(\frac{\alpha}{2}P\right)^{1/2}}{(1-\alpha)P}$$

$$P = 1$$

$$8\times 10^{-3} = \frac{\alpha^{3/2}}{\sqrt{2}}$$

$$\alpha^{3/2} = 8\sqrt{2}\times 10^{-3}$$

$$\alpha^3 = 128\times 10^{-6}$$

$$\alpha = \sqrt[3]{128}\times 10^{-2}$$

$$= 5.03\times 10^{-2}$$

$$= 5\times 10^{-2}$$

50. Correct answer is [10].

$Cr_2O_{7(aq)}^{-2} + 14\,H^+_{(aq)} + 6e^- \rightarrow 2\,Cr^{+3}_{(aq)} + 7H_2O(\ell)$

$$E_R = E_R^0 - \frac{0.059}{6}\log\frac{[Cr^{+3}]^2}{[Cr_2O_7^{-2}][H^+]^{14}}$$

$$0 = 1.33 - \frac{0.059}{6}\log\frac{10^{-6}}{[H^+]^{14}}$$

$$\frac{1.33\times 6}{0.059} = \log\frac{10^{-6}}{[H]^{14}}$$

$$135.254 = -6 - 1\;\; 4\log[H^+]$$

$$141.254 = 14\,\text{pH}$$

$$\text{pH} = \frac{141.254}{14} = 10.$$

Mathematics

Section A

51. Option (1) is correct.

Given: $L_1 : \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$

$L_2 : \frac{x-\lambda}{3} = \frac{y-4}{4} = \frac{z-5}{5}$

Let point on L_1 is A = (1, 2, 3)

point on L_2 is B = (λ, 4, 5)

Line L_1 is parallel to vector $\vec{p} = 2\hat{i}+3\hat{j}+4\hat{k}$

Line L_2 is parallel to vector $\vec{q} = 3\hat{i}+4\hat{j}+5\hat{k}$

$$\therefore \quad \vec{p}\times\vec{q} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix}$$

$$\vec{p}\times\vec{q} = -\hat{i}+2\hat{j}-\hat{k}$$

Shortest distance between L_1 and L_2

$$= \left|\frac{\overrightarrow{AB}\cdot\vec{p}\times\vec{q}}{|\vec{p}\times\vec{q}|}\right|$$

$$\frac{1}{\sqrt{6}} = \left|\frac{(\lambda-1)\hat{i}+2\hat{j}+2\hat{k}.(-\hat{i}+2\hat{j}-\hat{k})}{\sqrt{6}}\right|$$

$$|\lambda-3| = 1$$

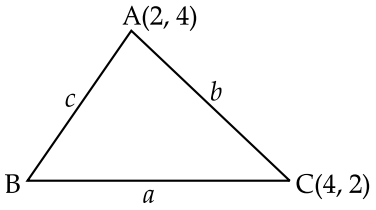
$$\lambda-3 = \pm 1$$

$$\lambda = 4, 2$$

Hence, Let $\lambda_1 = 4$,

$\lambda_2 = 2$

Circle passes through points (0, 0), (4, 2), (2, 4)



$a = \sqrt{20}$

$b = \sqrt{8}$

$c = \sqrt{20}$

$$D = \frac{1}{2}\begin{vmatrix} 1 & 1 & 1 \\ 0 & 4 & 2 \\ 0 & 2 & 4 \end{vmatrix}$$

$$D = \frac{12}{2} = 6$$

Radius of circle $= \frac{abc}{4\Delta}$

$$= \frac{\sqrt{20}\times\sqrt{20}\times\sqrt{8}}{4\times 6}$$

$$= \frac{5\times 2\sqrt{2}}{6}$$

$$= \frac{5\sqrt{2}}{3}$$

52. Option (2) is correct.

α is a root of $x^2 + x + 1 = 0$

$\alpha^2 + \alpha + 1 = 0$

$\alpha = \omega, \omega^2$ (Cube root of unity)

$$[4 \quad a \quad b]\begin{vmatrix} 1 & 16 & 13 \\ -1 & -1 & 2 \\ -2 & -14 & -8 \end{vmatrix} = [0 \quad 0 \quad 0]$$

$$[4-a-2b \quad 64-a-14b \quad 52+2a-56] = [0 \quad 0 \quad 0]$$

$\therefore \quad 4-a-2b = 0$

$64-a-14b = 0$

$52+2a-8b = 0$

$a+2b = 4$ ... (i)

$a+14b = 64$ ...(ii)

Solving Eqs. (i) and (ii), we get

$a = -6,$

$b = 5$

$$\therefore \frac{4}{\alpha^4}+\frac{m}{\alpha^{-6}}+\frac{n}{\alpha^5} = 3$$

$$\Rightarrow \quad \frac{4}{\omega}+\frac{m}{1}+\frac{n}{\omega^2} = 3$$

$$\Rightarrow 4\omega^2 + m + n\omega = 3$$

$$\Rightarrow 4\left(-\frac{1}{2}-\frac{\sqrt{3}}{2}i\right)+m+n\left(-\frac{1}{2}+\frac{\sqrt{3}}{2}i\right) = 3$$

$$\left(-2+m-\frac{n}{2}\right)+i\left(-2\sqrt{3}+\frac{n\sqrt{3}}{2}\right) = 3 + oi$$

Comparing real and imaginary part both sides, we get

$$\therefore \quad -2+m-\frac{n}{2} = 3$$

$$\text{and } \frac{-4\sqrt{3}}{2}+\frac{n\sqrt{3}}{2} = 0$$

$$\text{Hence, } n = 4, m = 7$$

$$\therefore \quad m+n = 7+4$$

$$= 11$$

53. Option (4) is correct.

$$f(x) = \frac{x}{3}+\frac{3}{x}+3, x \neq 0$$

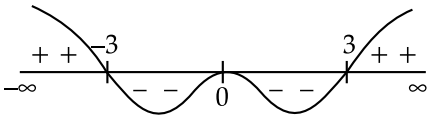
$$f(x) = \frac{1}{3}-\frac{3}{x^2} = 0$$

$$\Rightarrow \quad x = \pm 3$$

$$f(x) = \frac{x^2-9}{3x^2}$$

If $f(x)$ is increasing, then $f(x) > 0$

$$\Rightarrow \frac{x^2-9}{3x^2} > 0$$



$\therefore x \in (-\infty, -3) \cup (3, \infty)$

If $f(x)$ is decreasing then $f'(x) < 0$

$\therefore x \in (-3, 0) \cup (0, 3)$

$$\sum_{i=l}^{5} a_i^2 = (-3)^2 + (3)^2 + (-3)^2 + (0)^2 + (3)^2$$

$$= 9 + 9 + 9 + 9$$

$$= 36$$

54. Option (1) is correct.

$$P(\text{A}) = 0.7$$

$$P(\text{B}) = 0.4$$

$$P(\text{A} \cup \overline{\text{B}}) = 0.5$$

$$P\left(\frac{\text{B}}{\text{A} \cup \overline{\text{B}}}\right) = \frac{P\left(\text{B} \cap (\text{A} \cup \overline{\text{B}})\right)}{P\left(\text{A} \cup \overline{\text{B}}\right)}$$

$$= \frac{P\left((\text{B} \cap \overline{\text{B}}) \cup (\text{B} \cap \text{A})\right)}{P\left(\text{A} \cup \overline{\text{B}}\right)}$$

$$= \frac{P(\text{A} \cap \text{B})}{P\left(\text{A} \cup \overline{\text{B}}\right)}$$

$$= \frac{P(\text{A}) - P(\text{A} \cap \overline{\text{B}})}{P(\text{A}) + P(\overline{\text{B}}) - P\left(\text{A} \cap \overline{\text{B}}\right)}$$

$$= \frac{0.7-0.5}{0.7+0.6-0.5} = \frac{0.2}{0.8}$$

$$= \frac{1}{4}$$

55. Option (3) is correct.

$$\text{If } \frac{1}{1^4}+\frac{1}{2^4}+\frac{1}{3^4}+.....\infty = \frac{\pi^4}{90} \quad \text{... (i)}$$

$$\beta = \frac{1}{2^4}+\frac{1}{4^4}+\frac{1}{6^4}+\infty,$$

Taking $\frac{1}{16}$ common,

$$\beta = \frac{1}{16}\left\{\frac{1}{1^4}+\frac{1}{2^4}+\frac{1}{3^4}+...\right\}$$

$$= \frac{1}{16} \times \frac{\pi^4}{90}$$

$$\alpha = \frac{1}{1^4}+\frac{1}{3^4}+\frac{1}{5^4}+...$$

$$= \left(\frac{1}{1^4}+\frac{1}{2^4}+\frac{1}{3^4}+\frac{1}{4^4}+...\right) - \left(\frac{1}{2^4}+\frac{1}{4^4}+...\right)$$

$$\alpha = \frac{\pi^4}{90} - \frac{1}{16} \times \frac{\pi^4}{90}$$

$$= \frac{\pi^4}{90}\left(1-\frac{1}{16}\right)$$

$$= \frac{15}{16} \times \frac{\pi^4}{90}$$

$$\therefore \quad \frac{\alpha}{\beta} = \frac{\frac{15}{16} \times \frac{\pi^4}{90}}{\frac{1}{16} \times \frac{\pi^4}{90}}$$

$$= 15$$

56. Option (2) is correct.

Now, $x^2 - 2|x-3| - 5 = 0$

Case I Let $x - 3 \geq 0, x \geq 3$

$$x^2 - 2(x-3) - 5 = 0$$

$$x^2 - 2x + 1 = 0$$

$$x = 1, 1$$

But $x \geq 3$

$\Rightarrow \quad x \in \phi$

Case II $x - 3 < 0$

$x < 3$

$$x^2 + 2(x-3) - 5 = 0$$

$$x^2 + 2x - 11 = 0$$

$$x = \frac{-2 \pm \sqrt{4-4(-11)}}{2}$$

$$= -1 \pm \sqrt{12}, -1 - \sqrt{12}$$

Both values of $x < 3$, therefore both possible

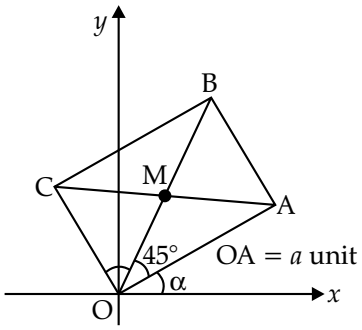
Sum of square of roots

$$= 3^2 + 1^2 + \left(-1+\sqrt{12}\right)^2 + \left(-1-\sqrt{12}\right)^2$$

$= 9+1+1+12-2\sqrt{12}+1+12+2\sqrt{12}$

$= 36$

57. Option (1) is correct.



$(\sqrt{3}+1)x+(\sqrt{3-1})y = 0$

$\therefore$ Slope of Line OB $= \dfrac{\sqrt{3}+1}{1-\sqrt{3}}$

$= \tan(105°)$

$\therefore \quad \alpha + 45° = 105°$

$\alpha = 60°$

point A= (a cos 60°, a sin 60°)

$= \left(\dfrac{a}{2}, \dfrac{a\sqrt{3}}{2}\right)$

A lie on $(\sqrt{3}-1)x-(\sqrt{3}+1)y+8\sqrt{3} = 0$

$\therefore \left(\dfrac{\sqrt{3}-1}{2}\right)a-\left(\dfrac{\sqrt{3}+1}{2}\right).\sqrt{3a}+8\sqrt{3} = 0$

$a\left[\dfrac{\sqrt{3}-1-3\sqrt{3}}{2}\right] = -8\sqrt{3}$

$a\left(\dfrac{-4}{2}\right) = -8\sqrt{3}$

$a = 4\sqrt{3}$

$a^2 = (4\sqrt{3})^2$

$= 48$

58. Option (4) is correct.

$\int_{\frac{1}{2}}^{1} 2xf(2x(1-2x))dx$

Let $t = 2x$

$\dfrac{dt}{dx} = 2$

If $x = -\dfrac{1}{2}$

$\Rightarrow \quad t = -1$

$x = 1$

$\Rightarrow \quad t = 2$

$\Rightarrow \quad I_1 = \dfrac{1}{2}\int_{-1}^{2} tf(t(1-t)dt)$

Now, using property $x \to a + b - x$

$I_1 = \dfrac{1}{2}\int_{-1}^{2}\{(l-t)f(l-t)(1-(1-t))\}dt$

$2I_1 = \int_{-1}^{2}(1-t)f(t(1-t))dt$

$\Rightarrow \quad 2I_1 = \int_{-1}^{2} f(t(1-t)dt - \int_{-1}^{2} tf(t)(1-t))dt$

$\Rightarrow \quad 2l_1 = I_2 - 2I_1$

$\Rightarrow \quad 4I_1 = I_2$

$\Rightarrow \quad \dfrac{I_2}{I_1} = 4$

59. Option (4) is correct.

Let vector $\vec{c}$ in plane of $\vec{a}$ and $\vec{b} = \lambda(\vec{a}+t\vec{b})$

$\vec{c}$ is perpendicular to $\vec{a}$.

$\vec{c}\cdot\vec{a} = 0$

$\lambda(\vec{a}+t\vec{b}).\vec{a} = 0$

$\vec{a}.\vec{a}+t(\vec{a}.\vec{b}) = 0$

$(\hat{i}+2\hat{j}+\hat{k}).(\hat{i}+2\hat{j}+\hat{k})+t(\hat{i}+2\hat{j}+\hat{k}).(2\hat{i}+\hat{j}+\hat{k}) = 0$

$6 + t(2 + 2 - 1) = 0$

$t = \dfrac{-6}{3} = -2$

$\therefore \quad \vec{c} = (\vec{a}-2\vec{b}).\lambda$

$= \lambda(\hat{i}+2\hat{j}+\hat{k}-4\hat{i}-2\hat{j}+2\hat{k})$

$\vec{c} = \lambda(-3\hat{i}+3\hat{k})$

$\vec{c} = \pm\dfrac{(\hat{i}-\hat{k})}{\sqrt{2}}$

60. Option (3) is correct.

$$x^2 + y^2 - 2x - 4y - 11 = 0$$

$$\text{centre of circle} = (1, 2)$$

$$\text{radius} = \sqrt{1+4+11}$$

$$= 4$$

$$3x^2 + py^2 = 4 \text{ passes through } (1, 2)$$

$$3 + p \times 4 = 4$$

$$p = \frac{1}{4}$$

$$\therefore \quad \text{Ellipse : } 3x^2 + \frac{y^2}{4} = 4$$

$$\frac{x^2}{4/3} + \frac{y^2}{16} = 1$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$a = \frac{2}{\sqrt{3}}, b = 4$$

$$b > a$$

$$1 - e^2 = \frac{a^2}{b^2}$$

$$e^2 = 1 - \frac{4}{3\times 16}$$

$$e = \sqrt{\frac{11}{12}}$$

Focal distance of the point C is $b \pm ek$

$$\text{where C} = (h, k)$$

$$f_1 = 4 + \sqrt{\frac{11}{12}} \times 2$$

$$f_2 = 4 - \sqrt{\frac{11}{12}} \times 2$$

$$\therefore \quad f_1 f_2 = 16 - \frac{11}{3} = \frac{37}{3}$$

$$\therefore \quad 6f_1 f_2 - r = 74 - 4 = 70$$

61. Option (3) is correct.

$$\text{Let, I} = \pi^2 \int_{-1}^{3/2} |x \sin \pi x| \, dx$$

$$= \pi^2 \left\{ \int_{-1}^{1} |x \sin \pi x| \, dx - \int_{1}^{3/2} x \sin \pi x dx \right\}$$

$$= \pi^2 \left\{ 2\int_{0}^{1} x \sin \pi x dx - \int_{1}^{3/2} x \sin \pi x dx \right\}$$

$$\text{Let, } I_1 = \int x \sin \pi x dx$$

Applying Integration by part's we get,

$$I_1 = \int x \sin \pi x dx$$

$$-\frac{1}{x}\cos \pi x + \int \frac{1}{\pi} \cos \pi x dx$$

$$= -\frac{x}{\pi}\cos \pi x + \frac{\sin \pi x}{\pi^2}$$

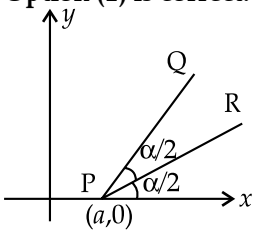
$$\text{I} = \pi^2 \left\{ \begin{matrix} 2\left(-\frac{x}{\pi}\cos \pi x + \frac{\sin \pi x}{\pi^2} \right)_0 \\ -\left(\frac{x}{\pi}\cos \pi x + \frac{\sin \pi x}{\pi^2} \right)_1^{3/2} \end{matrix} \right\}$$

$$= \pi^2 \left\{ \frac{2}{\pi} - \left(\frac{1}{\pi^2} - \frac{1}{\pi} \right) \right\}$$

$$= \pi^2 \left\{ \frac{3}{\pi} + \frac{1}{\pi^2} \right\}$$

$$= 3\pi + 1$$

62. Option (1) is correct.



$$\text{Slope of Line PR} = 2 - \sqrt{3}$$

$$\tan\left(\frac{\alpha}{2}\right) = 2 - \sqrt{3}$$

$$\therefore \quad \frac{\alpha}{2} = 15°$$

$$\therefore \quad \alpha = 30°$$

Equation of line PR,

$$y = \left(2 - \sqrt{3}\right)(x - a)$$

Distance of line PR from origin is $\frac{1}{\sqrt{2}}$

$$\frac{1}{\sqrt{2}} = \left| \frac{\sqrt{3}a - 2a}{\sqrt{4+3-4\sqrt{3}+1}} \right|$$

$$\frac{1}{\sqrt{2}} = \left| \frac{|a|(\sqrt{3}-2)}{2\sqrt{2-\sqrt{3}}} \right|$$

$$\sqrt{2} = |a|\sqrt{2-\sqrt{3}}$$

$$2 = a^2\left(2 - \sqrt{3}\right)$$

$$\therefore \quad 3a^2\tan\alpha - 2\sqrt{3} = \frac{3\times 2}{2-\sqrt{3}} \times \frac{1}{3} - 2\sqrt{3}$$

$$= \frac{2}{2-\sqrt{3}} - 2\sqrt{3}$$

$$= \frac{2(2+\sqrt{3})}{(2-\sqrt{3})(2+\sqrt{3})} - 2\sqrt{3}$$

$$= \frac{4+2\sqrt{3}}{4-3} - 2\sqrt{3}$$

$$= 4$$

63. Option (4) is correct.

Total number of triangles $= {}^{n}C_3 - {}^{m}C_3$

n = all possible points

m = collinear points

$${}^{12}C_3 - {}^{5}C_3 = 210$$

64. Option (1) is correct.

Let $z = \dfrac{2\cos\theta + i\sin\theta}{\cos\theta - 3i\sin\theta}$

We know that, $z + \bar{z} = 2\text{Re}(z)$

$$\text{Re}\left(\frac{2\cos\theta + i\sin\theta}{\cos\theta - 3i\sin\theta}\right) = -\frac{1}{10}$$

$$\frac{2\cos\theta + i\sin\theta}{\cos\theta - 3i\sin\theta} + \frac{2\cos\theta - i\sin\theta}{\cos\theta + 3i\sin\theta} = 2\times\left(\frac{-1}{10}\right)$$

$$\Rightarrow \frac{(2\cos^2\theta - 3\sin^2\theta) + (2\cos^2\theta) - (3\sin^2\theta)}{\cos^2\theta + 9\sin^2\theta}$$

$$= \frac{-2}{10}$$

$$\Rightarrow \frac{2\cos^2\theta - 3\sin^2\theta}{\cos^2\theta + 9\sin^2\theta} = \frac{-1}{10}$$

$$\Rightarrow 20\cos^2\theta - 30\sin^2\theta = -\cos^2\theta - 9\sin^2\theta$$

$$\Rightarrow 21\cos^2\theta - 21\sin^2\theta = 0$$

$$\cos^2\theta = 0$$

$$20 = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}$$

$$\Rightarrow \Sigma\theta^2 = \frac{\pi^2}{16} + \frac{9\pi^2}{16} + \frac{25\pi^2}{16} + \frac{49\pi^2}{16}$$

$$= \frac{27\pi^2}{4}$$

65. Option (3) is correct.

$\max\{x, y\} = \{3, 4\}$

$A = \{0, 1, 2, 3, 4, 5\}$

$R = \{(0, 3), (3, 0), (0, 4), (4, 0) (1, 3), (3, 1), (1, 4), (4, 1), (2, 3), (3, 2), (2, 4), (4, 2), (3, 3), (3, 4), (4, 3), (4, 4)\}$

Total 16 elements

Not reflexive as (0, 0), ..., $(2, 2) \notin R$

R is symmetric, if (a, b) and $(b, a) \in R$, V all a, b above condition is true. Hence symmetric R is transitive if (a, b) and $(b, c) \in R$, then $(a, c) \in R$.

$(a, c) \in R$

$(0, 3), (3, 1) \in R$

But, $(0, 1) \notin R$

$\therefore$ R is not transitive.

$\Rightarrow$ only S_2 is true.

66. Option (4) is correct.

$T_{r+1} = {}^{1016}C_r\ (5)^{\frac{1016-r}{2}}.7^{\frac{r}{8}}$ $\dfrac{1016-r}{2}$, $\dfrac{r}{8}$ both must be integer,

r	$\frac{r}{8}$	$\frac{1016-r}{2}$
0	I	I
8	I	I
16	I	I
:		
1016	I	I

$$1016 = 0 + (n-1)8$$

$$\Rightarrow n - 1 = \frac{1016}{8} = 127$$

So, $n = 128$

67. Option (3) is correct.

$$f(x) = x - 1$$

$$f(f(x)) = f(x) - 1 = x - 1 - 1 = x - 2$$

$$g(f(f(x))) = e^{x-2}$$

$$\therefore \frac{dy}{dx} = e^{2\sqrt{x}} \times e^{x-2} - \frac{1}{\sqrt{x}}y$$

This is linear different equation

$$\frac{dy}{dx} + \frac{1}{\sqrt{x}}y = e^{x-2\sqrt{x}-2}$$

$$\frac{dy}{dx} + P(x)y = Q(x)$$

$$\therefore P(x) = \frac{1}{\sqrt{x}},\ Q(x) = e^{x-2\sqrt{x}-2}$$

$$\text{I.F.} = e^{\int\frac{dx}{\sqrt{2}}} = e^{2\sqrt{x}}$$

Its solution is

$$y \times e^{2\sqrt{x}} = \int e^{2\sqrt{x}} \times e^{x-2\sqrt{x}-2}dx + c$$

$$y \times e^{2\sqrt{x}} = \int e^{x-2}dx + c$$

$y \times e^{2\sqrt{x}} = e^{x-2} + c$

Given, $x = 0, y = 0$

$\Rightarrow \quad 0 = e^{-2} + c;$

$c = -e^{-2}$

$\therefore \quad y \times e^{2\sqrt{x}} = e^{x-2} + c$

Given, $x = 0, y = 0$

$\Rightarrow e^{-2} + c$

$c = -e^{-2}$

$\therefore \quad y \times e^{2\sqrt{x}} = e^{x-2} - e^{-2}$

when $x = 1, y \times e^2 = e^{-1} - e^{-2}$

$$y = \frac{e^{-1} - e^{-2}}{e^2} = \frac{\frac{1}{e} - \frac{1}{e^2}}{e^2} = \frac{e^2 - e}{e^5} = \frac{e-1}{e^4}$$

68. Option (1) is correct.

$$\cot^{-1}\left(\frac{\sqrt{1+\tan^2 2}-1}{\tan 2}\right) - \cot^{-1}\left(\frac{\sqrt{1+\tan^2\left(\frac{1}{2}\right)}+1}{\tan\left(\frac{1}{2}\right)}\right)$$

we know that, $1 + \tan^2\theta = \sec^2\theta$

$$\Rightarrow \cot^{-1}\left(\frac{|\sec 2|-1}{\tan 2}\right) - \cot^{-1}\left(\frac{\left|\sec\frac{1}{2}\right|+1}{\tan\frac{1}{2}}\right)$$

$(\therefore 1 + \cos 2\theta = 2\cos^2\theta)$

$$= \cot^{-1}\left(\frac{-2\cos^2 1}{2\sin|\cos|}\right) - \cot^{-1}\left(\frac{2\cos^2\left(\frac{1}{4}\right)}{2\sin\left(\frac{1}{4}\right)\cos\left(\frac{1}{4}\right)}\right)$$

$$= \cot^{-1}(-\cot 1) - \cot^{-1}\left(\cot\frac{1}{4}\right)$$

$$= \pi - \cot^{-1}(\cot 1) - \cot^{-1}\left(\cot\frac{1}{4}\right)$$

$$= \pi - 1 - \frac{1}{4} = \frac{5}{4}$$

69. Option (2) is correct.

$$A = \begin{bmatrix} 2 & 2+p & 2+p+q \\ 4 & 6+2p & 8+3p+2q \\ 6 & 12+3p & 20+6p+3q \end{bmatrix}$$

Now taking determinant of both side,

$$A = \begin{vmatrix} 2 & 2+p & 2+p+q \\ 4 & 6+2p & 8+3p+2q \\ 6 & 12+3p & 20+6p+3q \end{vmatrix}$$

$C_3 \to C_3 \to C_2 - C_1 \times \frac{q}{2}$

Then $C_3 \to C_2 - C_1 x\left(1+\frac{p}{2}\right)$

$$\Rightarrow |A| = \begin{vmatrix} 2 & 0 & 0 \\ 4 & 2 & 2+p \\ 6 & 6 & 8+3p \end{vmatrix}$$

$|A| = 2\,(16 + 6p - 12 - 6P)$

$= 8$

$\det(ab)\,(\text{adj})(3A))) = 2^m.3^n$

$\therefore \; |\text{adj}\,(\text{adj})(3A))| = |3A|^{(3-1)2} = |3A|^{-4}$

$= (3^3\,|A|)^4 = (3^3 \times 2^3)^4 = 2^{12} \times 3^{12}$

$\Rightarrow \quad m + n = 24$

70. Option (4) is correct.

$$\lim_{x\to 0}\left|\frac{\tan^{-1}x + \frac{1}{2}[\ln(1+x) - \ln(1-x) - 2x]}{x^5}\right|$$

Applying senes expansion,

$$\tan^{-1}x = x - \frac{x^3}{3} + \frac{x^5}{5}.....$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3}.....$$

$$\ln(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3}.....$$

$$= \lim_{x\to 0}\frac{\left(x - \frac{x^3}{3} + \frac{x^5}{5}....\right) + \frac{1}{2}\left[x - \frac{x^2}{2} + \frac{x^3}{3}... - \left(-x - \frac{x^2}{2} - \frac{x^3}{3}...\right)\right] - 2x}{x^5}$$

$$= \lim_{x\to 0}\frac{2x + \frac{2x^5}{5}... - 2x}{x^5} = \frac{2}{5}$$

$$\lim_{x\to 1}(x)^{\frac{2}{1-x}}$$

A is $x \to 1, \; \frac{2}{1-x} \to \infty$

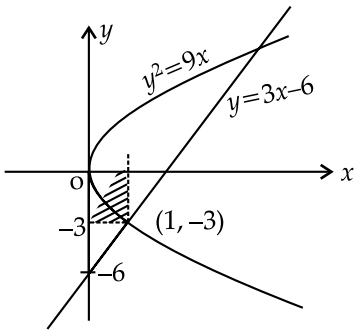
$$\lim_{x\to1} x^{\frac{2}{(1-x)}} = e \lim_{x\to1} x^{\frac{2}{(1-x)}(x-1)} = e^{-2}$$

$\Rightarrow$ Both statement are correct

Section B

71. Correct answer is [15].

$0 \le 9x \le y^2$ and $y \ge 3x - 6$



A= Required Area

$$= \int_0^1 (-3\sqrt{x} - 3x + 6)dx$$

$$\left\{\because \int x^n dx = \frac{x^{n+1}}{n+1}\right\}$$

$$A = -3\left(\frac{x^{\frac{3}{2}}}{\frac{3}{2}}\right)\Bigg|_0^1 - \left(\frac{3x^2}{2} - 6x\right)\Bigg|_0^1$$

$$A = -2[1-0]\left[\frac{3}{2} - 6\right]$$

$$A = -2\frac{3}{2} + 6 = \frac{5}{2} \text{ sq unit}$$

$$\therefore \quad 6A = 6 \times \frac{5}{2} = 15$$

72. Correct answer is [96].

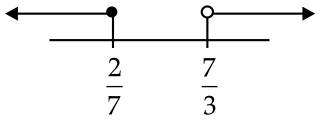
$$f(x) = \cos^{-1}\left(\frac{4x+5}{3x-7}\right)$$

$$\Rightarrow -1 \le \left(\frac{4x+5}{3x-7}\right) \le 1$$

$$\left(\frac{4x+5}{3x-7}\right) \ge -1 \quad \{-\le t \le 1, 9f(y) = \cos^{-1} t\}$$

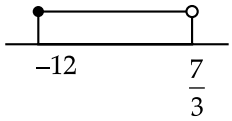
$$\frac{4x+5+3x-7}{3x-7} \ge 0$$

$$\Rightarrow \frac{7x-2}{3x-7} \ge 0$$



$$x \in \left(-\infty, \frac{2}{7}\right) \cup \left(\frac{7}{3}, \infty\right)$$

and $\frac{4x+5}{3x-7} \le 1 \Rightarrow \frac{x+12}{3x-7} \le 0$



$\therefore$ Domain of $f(x)$ is $= \left[-12, \frac{2}{7}\right]$

$$\alpha = -12, \beta = \frac{2}{7}$$

$g(x) = \log_2(2 - 6\log_{27}(2x + 5))$ Domain

$2 - 6\log_{27}(2x + 5) > 0$

$\Rightarrow 6\log_{27}(2x + 5) < 2$

$\Rightarrow \log_{27}(2x + 5) < \frac{1}{3}$

$\Rightarrow 2x + 5 < 3$

$\Rightarrow x < -1$

and $2x + 5 > 0$

$\Rightarrow x > -\frac{5}{2}$

Domain is $= \left(-\frac{5}{2}, -1\right)$

$$\gamma = \left(-\frac{5}{2}, \delta = 1-\right)$$

$$|7(\alpha+\beta) + 4(\gamma+\delta)| = \left|7\left(-12+\frac{2}{7}\right) + 4\left(-\frac{5}{2}-1\right)\right|$$

$$|-82 - 14| = 96$$

73. Correct answer is [56].

$L_1 : x + 2 = y - 1 = z = \ell$

$$L_2 : \frac{x-3}{5} = \frac{y}{-1} = \frac{z-1}{1} = m$$

$$L_3 : \frac{x}{-3} = \frac{y-3}{3} = \frac{z-2}{1} = n$$

any point on L_1 ; $l - 2, l + 1, l$

any point on L_2 ; $5m + 3, -m, m + 1$

any point on L_3 ; $-3n, 3n + 3, n + 2$

Point of intersection of L_1 and L_2

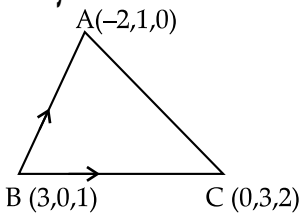
$$\left.\begin{aligned} \ell - 2 &= 5m + 3 \\ \ell + 1 &= -m \\ \ell &= m + 1 \end{aligned}\right\} \ell = 0, m = -1 \text{ A } (-2, 1, 0)$$

Point of intersection of L_2 and L_3

$$\left.\begin{aligned} 5m + 3 &= -3n \\ -m &= 3n + 3 \\ m + 1 &= n + 2 \end{aligned}\right\} m = 0,\ n = -1 \text{ B } (3, 0, 1)$$

Point of intersection L_3 and L_4

$$\left.\begin{aligned} -3n &= \ell - 2 \\ 3n + 3 &= \ell + 1 \\ n + 2 &= \ell \end{aligned}\right\} \ell = 2,\ n = 0 \text{ C } (0, 3, 2)$$



$$\text{Area } (\Delta ABC) = \frac{1}{2}|\overrightarrow{AB} \times \overrightarrow{AC}|$$

$$\text{Ar}(\Delta ABC) = \left|\frac{1}{2}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -5 & 1 & -1 \\ -3 & 3 & 1 \end{vmatrix}\right|$$

$$A = \frac{1}{2}\left|\hat{i}(4) - \hat{j}(-8) + \hat{k}(-12)\right|$$

$$A = \frac{1}{2}\sqrt{16 + 64 + 144} = \sqrt{56}$$

$$A^2 = 56$$

74. Correct answer is [63].

$(1919)^{1919} = (1920 - 1)^{1919}$

$= {}^{1919}C_0(1920) - {}^{1919}C_1(1920)^{1918} + \ldots.$

$+ {}^{1919}C_{1918}(1920)^1 - {}^{1919}C_{1919}$

$= 100t + 1919 \times 1920 - 1$

$= 100t + (1920 - 1) \times 1920 - 1$

$= 100t + (1920)^2 - 1920 - 1$

$= 100t + (1920)^2 - 1921$

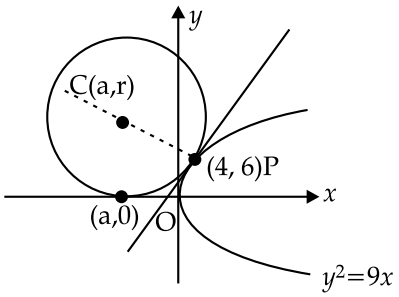
$= 100t + 2000 - 1921$ (Combining values)

$= 100\lambda + 79$

$\Rightarrow$ Number having last two digit is 79.

$\therefore$ Product of last two digit is 63.

75. Correct answer is [30].



Equation of circle,

$(x - a)^2 + (y - r)^2 = r^2$

Circle passes through point (4, 6)

$(4 - a)^2 + (6 - r)^2 = r^2$

$16 + a^2 - 8a + 36 + r^2 - 12r = r^2$

$a^2 - 8a - 12r + 52 = 0$ ... (i)

Tangents to parabola at (4, 6) is

$$6y = 9\left(\frac{x + 4}{2}\right)$$

$$4y = 3x + 12$$

$$3x - 4y + 12 = 0$$

This is also tangents to the circle

$\therefore$ $CP = r$

$$\frac{3a - 4r + 12}{5} = \pm r$$

$$3a - 4r + 12 = \pm 5r$$

$\therefore 3a + 12 = 9r$ or $3a + 12 = -r$ ...(ii)

From Eqs. (i) and (ii), we get

if $a + 4 = 3r$, then

$a^2 - 8a - 4(a + 4) + 52 = 0$

$a^2 - 12a + 36 = 0$

$a = 6$ (Rejected)

If $3a + 12 = -r$, then

$a^2 - 8a + 12(3a + 12) + 52 = 0$

$a^2 + 28a + 196 = 0$

$\therefore$ $a = -14$

$r = 30$

■■■